

O-RAN Open Fronthaul S-Plane Software Testbed

Configuration, Setup and Standards Reference

Document scope	Every hardware and software configuration used by the testbed, each traced to the standard or reference implementation that fixes it. No value in this document is assumed — all were read back from the running system or from the cited standard.
Telecom profile	ITU-T G.8275.1 — PTP telecom profile for phase/time distribution with <i>full timing support</i> from the network. Selected because O-RAN fronthaul sync configurations LLS-C1/C2/C3 distribute timing through the network, and because the real TIMESAFE hardware captures used for comparison were verified to run domain 24 (G.8275.1) on all 260,525 PTP packets.
Frozen state	15 artefacts SHA-256 hashed at 2026-09-17T01:21:41Z. <code>verify_frozen.sh</code> refuses to run if any changed.
Validation result	81 admissible runs (45 attack / 36 benign) across 9 replicates. Sensitivity 1.000 [0.921, 1.000]; specificity 1.000 [0.904, 1.000]; fault-attribution accuracy 0.956 [0.852, 0.988].

Reading rule. Constants are in two classes. **Class A — protocol legality:** discrete wire fields whose legal value is stated by a standard; a software testbed reproduces these exactly, so no calibration exists or is needed. **Class B — measured quantities:** offsets, delays, rates. Standards give these in nanoseconds, but this testbed's software-timestamping floor is microsecond-scale, so the standards' absolute limits are deliberately *not* applied; thresholds are expressed as multiples of a measured floor. Every table below marks which class a value belongs to.

1 Hardware / platform configuration

The testbed runs entirely in software. The table records the actual platform and, critically, what that platform *cannot* provide — each limit verified by direct query rather than assumption.

Item	Value (read from the system)	How verified	Consequence
Kernel	6.18.44-fc-v33 (Firecracker microVM)	<code>uname -r</code>	Monolithic build
Privileges	uid 0, CAP_NET_ADMIN present	<code>id -u; capsh --print</code>	Namespaces and bridges permitted
Loadable modules	None — no <code>/lib/modules</code>	<code>modprobe ptp_mock → Module not found</code>	ptp_mock / ptp_kvm cannot be loaded , so no per-node mock PHC
PTP hardware clock	None — no <code>/dev/ptp*</code>	<code>ls /dev/ptp*</code>	No hardware clock to discipline
veth timestamping	software-transmit, software-receive only; PTP Hardware Clock: none	<code>ethtool -T v-gma</code>	Microsecond noise floor vs a nanosecond target
Oscillator	One shared kernel clock from one physical crystal	Architectural — all namespaces share the host clock	Clocks cannot drift apart ; B1/B6 impossible

What a second physical machine would buy. Two machines means two crystals, so clocks genuinely drift apart — this alone makes **B6 (oscillator drift)** real. Adding an Intel i210/i226 NIC in each and a direct back-to-back cable adds hardware timestamping, dropping the floor from microseconds to tens of nanoseconds and making **A4** and **B5** measurable. It does *not* unlock B1, B4, A6 or A7 — those additionally require a GNSS timing receiver or a SyncE-capable PHY.

2 G.8275.1 profile configuration (Class A — exact)

This is the contents of `cfg/g87251.base`, applied to every node. Each row gives the value, the standard that fixes it, and — where applicable — the same field observed in the real TIMESAFE hardware capture, to show the testbed and the real data agree.

linuxptp key	Value	Standard / basis	Observed in real capture
domainNumber	24	G.8275.1 — telecom range 24–43, default 24	24 on all 260,525 PTP pkts ✓
logSyncInterval	-4 (16/s, 62.5 ms)	G.8275.1 mandated Sync rate	-4 on all 114,574 Sync ✓
logAnnounceInterval	-3 (8/s, 125 ms)	G.8275.1 mandated Announce rate	-3 on all 81,969 Announce ✓
logMinDelayReqInterval	-4 (16/s)	G.8275.1	0x7F = 'unspecified', legal per IEEE 1588 cl.13.3.2.14
announceReceiptTimeout	3	G.8275.1 typical (range 2–10)	—
priority1	128	G.8275.1 fixes it at 128 and <i>removes it from the BMCA comparison</i>	128 on legitimate clocks ✓

linuxptp key	Value	Standard / basis	Observed in real capture
delay_mechanism	E2E	G.8275.1 requires end-to-end	Delay_Req/Delay_Resp present ✓
network_transport	L2	G.8275.1 — Ethernet, no IP	EtherType 0x88F7 ✓
ptp_dst_mac	01:1B:19:00:00:00	G.8275.1 forwardable multicast group	100% conformant ✓
path_trace_enabled	1	IEEE 1588 PATH_TRACE TLV; G.8275.1 uses it for timing-loop prevention	Absent — 0 TLVs in 81,969 Announce
G.8275.*.localPriority	128	G.8275.1 alternate-BMCA port tiebreaker	—
maxStepsRemoved	255	IEEE 1588 / G.8275.1	—

Why this matters most. The project's earlier netem runs used `domain 0`, `logSync -3` and `logAnnounce +1` — one Announce every **2 seconds** against the standard's **125 ms**, a 16× mismatch. Because the benign data came from that testbed and the attack data from real G.8275.1 hardware, any classifier could have separated the two classes on message cadence alone and learned nothing about attacks. Aligning the testbed to G.8275.1 removes that confound. This is the single most important configuration decision in the document.

3 Declared deviations from a production deployment

Three settings differ from a real fronthaul. Each is stated with its reason; none is a workaround that hides a limitation.

Setting	Why it differs	What it costs
free_running 1	All namespaces share one kernel clock. With the servo active, grandmaster and clients would fight over the <i>same</i> clock — physically meaningless and it would corrupt host timekeeping.	Offsets are genuinely measured and logged but not applied. Servo convergence dynamics are absent.
time_stamping software	veth exposes no PHC and no hardware timestamp modes (ethtool -T).	Microsecond floor. No conformance claim against G.8273.2 or G.8271.1 is possible.
clock_class_thresho ld 248	This is an acceptance policy — the worst master a client will accept — not a quality claim. The lab has no PRTC/GPS, so every node truthfully advertises clockClass 248. At the default of 7, every node rejects every other and nothing synchronises.	No node is given a hand-set clockClass 6. Hierarchy is established with priority2, which is an operator-set field by design.

4 Topology and node roles

Two Ethernet bridges separated by a Boundary Clock. This models an O-RAN **LLS-C2/C3** fronthaul, where timing crosses shared switching between the O-DU/grandmaster site and the radio units — the configurations TIMESAFE identified as most exposed, because the timing path traverses equipment an attacker can reach.

[gma] [gmb] ■■■■ brUP ■■■■ [bc:v-bc-up | v-bc-dn] ■■■■ brDN ■■■■ [ru1] [ru2] [ru3]

Node	Role	priority2	Port config	Purpose in the experiment
gma	Primary grandmaster (T-GM)	100 (best)	serverOnly 1	Wins the BMCA under normal operation
gmb	Provisioned backup GM	110	serverOnly 1	Enables B2 — a legitimate, planned failover
bc	Boundary clock (T-BC)	120	v-bc-up client / v-bc-dn server	Makes stepsRemoved vary (0 upstream → 1 downstream); enables A8 and B7
ru1/ru2/ru3	O-RU proxies (T-TSC)	128	clientOnly 1	Three clients provide the spatial axis — a fault on one node vs all nodes under one GM

Why priority2 and not clockClass. Because nothing in this lab is GPS-traceable, fabricating a clockClass of 6 on any node would be inventing evidence. Instead the hierarchy uses priority2 — an operator-configured BMCA input whose whole purpose is expressing local preference. Every node continues to advertise its true, untraceable clockClass 248.

5 Software stack

Component	Version	Role
linuxptp (ptp4l, pmc, phc2sys)	4.0	Reference PTP implementation. All PTP behaviour is produced by real linuxptp — none is simulated.
iproute2	6.1.0	Namespaces, veth pairs, bridges, and netem impairment for B3
tcpdump / libpcap	4.99.4	Full-frame capture on both segments
scapy	installed	Frame injection for A2 (forged Sync) and A5 (flood)
tcpreplay	installed	Retransmission of captured frames for A3 (replay)
Python	3.11.15	Extraction, windowing, decision rule, evaluation

6 Processing pipeline and frozen artefacts

Stage	Artefact	SHA-256 (first 16)	Function
Topology	run/topology.sh	334f44f29bf8d5a	Builds 2 bridges, 6 namespaces, 7 veth ports
Launch	run/start.sh	cd736fbfbe38fa64	Interface-readiness barrier, then ptp4l per node + dual capture
Profile	cfg/g87251.base	afe3b8935354033e	The G.8275.1 configuration in §2
Scenarios	run/scenarios.sh	e27ad8aedd2af7f4	9 scenarios + per-run context manifest
Injection	run/inject.py / flood.py	8b438ef70981d267 / 9455405115809f95	Real forged frames on the wire (A2, A5)
Telemetry	run/pmc_log.sh	c872727a4f163c33	Polls 7 management datasets per node per second

Stage	Artefact	SHA-256 (first 16)	Function
Extraction	ptp_deep_extract.py	3e0fa9e9ba3a34bb	56 columns per packet, full IEEE 1588 surface
Windowing	run/window_features.py	46b2fetc3d41e748	0.4 s windows, 0.2 s step → detector input
Decision	run/decision_rule.py	99a7a29ae830cdb2	Frozen rule, no tunable parameter
Floor	run/baseline_floor.py	b9c78c1cfd1670c8	Instrument characterisation per ITU-T G.8260
Evaluation	run/evaluate.py	16175e19381d3159	Session-disjoint, Wilson CIs, explicit UNKNOWN

7 Every constant in the decision rule, and where it comes from

The rule contains no fitted parameter. Each constant is either fixed by a standard or is provisioned configuration read from the run's context manifest.

Constant	Value	Class	Source
Domain range	24–43	A — exact	ITU-T G.8275.1
priority1	128	A — exact	ITU-T G.8275.1 (fixed, removed from BMCA comparison)
Minimum legal clockClass	6	A — exact	IEEE 1588-2019 Table 5 — values 0–5 are RESERVED, so an Announce carrying them is illegal, not merely unusual
Announce rate	8/s (log -3)	A — exact	ITU-T G.8275.1
Destination MAC	01:1B:19:00:00:00	A — exact	ITU-T G.8275.1
sequenceId monotonicity	strictly increasing	A — exact	IEEE 1588. Caveat applied: Delay_Resp echoes the <i>requester's</i> sequenceId (cl.11.3), so the counter is keyed per requesting port
GM allow-list, BC identity, client inventory, maintenance flag	per run	Provisioned	Operator configuration in context.json — a provisioned fact, not a threshold
Transient fraction	1%	Structural	Definition of a start-up transient (a node announcing itself before it locks). Not fitted to observations
Cadence tolerance	2×	Structural	Tolerance for burstiness/duplication when comparing a source's observed rate against the rate it declares in its own logMessageInterval. Not fitted

8 Measured instrument floor (Class B denominator)

From 9 healthy baseline runs, 1,159 windows. Metric definitions follow **ITU-T G.8260**, the recommendation that defines packet-timing metrics and measurement methodology. These are properties of the *instrument*, not of a network, and support no conformance claim.

Metric	p50	p95	max
Inter-arrival mean (ms)	7.073	8.594	17.019
Inter-arrival std (ms)	11.097	13.924	25.064
Message rate (Hz)	144.109	168.899	247.492
Announce rate (Hz)	8.116	10.574	10.663
Sync rate (Hz)	17.110	18.642	61.873

The observed Announce rate of 8.116 Hz and Sync rate of 17.110 Hz confirm the testbed is transmitting at the G.8275.1 mandated 8/s and 16/s respectively.

9 Scenario configurations

ID	Scenario	How it is produced on the wire	O-RAN WG11 threat
—	baseline	All nodes nominal; GM-A grandmaster via BC to the three RUs	n/a
A1	Rogue grandmaster	A ptp4l instance with priority2=1, identity not on the allow-list, self-announcing on the RU segment	T-SPLANE-02 / 03
A2	Sync/Follow_Up spoof	scapy injects forged Sync with an implausible originTimestamp from an unprovisioned source; never sends Announce	T-SPLANE-02 family
A3	Replay	A live slice is captured then retransmitted with tcpreplay (stale timestamps, reused sequenceId)	cf. T-SPLANE-04
A5	DoS / flooding	scapy floods the RU segment with high-rate PTP frames	T-SPLANE-01
A8	Rogue boundary clock	A two-port ptp4l (priority2=115) slaves upstream on brUP and masters downstream on brDN, relaying the real grandmaster with stepsRemoved incremented	T-SPLANE-03
B2	Planned GM failover	GM-A withdrawn for maintenance; BMCA elects GM-B, which IS on the allow-list, with the maintenance window open	n/a — benign
B3	PDV / congestion	netem applies 5 ms ± 3 ms normal-distribution delay to RU links	n/a — benign
B7	Topology change	RU3's link is bounced and re-homed during an open maintenance window	n/a — benign

A1 vs B2 is the decisive pair. Both present the same surface symptom — a new grandmaster identity appears. They are separated only by provisioned context: the rogue's identity is absent from the allow-list, whereas the failover's is the configured backup during a declared maintenance window. Without that context layer the two are genuinely indistinguishable from the packets alone.

A8 design note. A8 was initially built as a single-port self-announcing master, which is structurally identical to A1 — the same fault implemented twice. It was rebuilt as a genuine two-port relaying boundary clock. The discriminator is now principled: a rogue grandmaster announces *itself* with stepsRemoved 0, whereas a rogue boundary clock *relays* another clock's grandmaster identity with stepsRemoved incremented.

10 Faults deliberately out of scope, with the physical reason

Fault	Why it cannot be produced or measured here	What would be required
A6 — GNSS spoofing	No GNSS receiver exists in software. Fabricating a gnss-sync-status value would be inventing evidence, not producing it.	GNSS simulator + receiver + RF-shielded chamber (transmitting these signals over the air is illegal in most jurisdictions)
A7 — GNSS jamming	Requires real RF denial against a real receiver (C/N0, AGC) and a real oscillator to drift.	As above, plus an independent oscillator
B1 — GNSS holdover	Requires a genuine reference to lose and an oscillator with a characterised holdover curve.	GNSS timing receiver with PPS + independent oscillator. A second PC alone is insufficient.
B4 — SyncE / EEC	SyncE is a physical-layer function; veth has no PHY, therefore no ESMC, no SSM and no EEC.	SyncE-capable NIC and switch (Intel E810 class) with syncE4l
B6 — Oscillator drift	All namespaces share one kernel clock from one crystal; nothing can drift relative to anything.	A second physical machine — this is the one a second PC solves
A4 — Delay attack	netem can inject the asymmetry, but a passive capture yields only t1 and t4; the client's t2/t3 never appear on the wire.	Live pmc telemetry per client (now logged), or a redundant second path; hardware timestamping for any ns-scale claim

11 Experimental method and validity controls

Control	How it is enforced
No tunable parameters	Every constant in the decision rule is a standard, a provisioned fact, or a structural definition — enumerated in §7.
Pre-registration	<code>freeze.sh</code> SHA-256 hashes all 15 artefacts into <code>FROZEN.json</code> . <code>verify_frozen.sh</code> was run and passed before every admissible replicate.
Development/validation separation	Replicates 1–13 and 20–21 exposed defects, were used to fix them, and are excluded and deleted. Replicates 30–38 were produced after the final freeze with the rule untouched.
Session-disjoint splitting	Evaluation is by run, never by window; no window from a run appears on both sides.
Fail-closed	Missing or unreadable evidence yields UNKNOWN, never BENIGN. An infrastructure failure (empty capture) is detected, retried once, and discarded with disclosure rather than scored as a detection result.
Underpowered reporting	If a class is too small the evaluator reports NOT ESTABLISHED with the interval width and both class sizes — never a bare pass/fail.

12 Validation result

Metric	Value	95% Wilson CI	n
Sensitivity (attack recall)	1.000	[0.921, 1.000]	45
Specificity (benign recall)	1.000	[0.904, 1.000]	36
Fault-attribution accuracy	0.956	[0.852, 0.988]	45

Per-fault attribution: $A1 = 9/9$ · $A2 = 9/9$ · $A3 = 8/9$ · $A5 = 9/9$ · $A8 = 8/9$. Confusion — attack: TP 45, FN 0, UNKNOWN 0; benign: TN 36, FP 0, UNKNOWN 0.

The two attribution misses are genuine signature overlap, not defects. In replicate 30 a replay was attributed to A5 because looping a captured slice genuinely inflated the Announce rate to 22.4 Hz against the 8 Hz the source itself declares — a replay executed at speed *is* a form of flooding. In the same replicate a rogue boundary clock was attributed to A3 because inserting it caused downstream clients to re-parent, producing genuine sequence restarts. The binary attack/benign verdict was correct in both cases. These were left uncorrected deliberately: re-ordering the rule to chase 100% after observing the result would have converted validation data into development data and destroyed the very property that makes the figure meaningful.

13 Standards and references

Reference	Used for
ITU-T G.8275.1	The telecom profile: domain 24–43, Sync 16/s, Announce 8/s, priority1 fixed 128, L2 multicast, alternate BMCA with localPriority and masterOnly
IEEE 1588-2019	Message and dataset structure; Table 5 clockClass reservations; cl.11.3 Delay_Resp sequenceId semantics; cl.13.3.2.14 logMessageInterval 0x7F
RFC 8173	Complete enumeration of defaultDS / currentDS / parentDS / timePropertiesDS / portDS fields (open IETF source, used instead of recalling field names)
ITU-T G.8271.1	Network time-error budget; PRTC-to-application limit of 1.5 µs
ITU-T G.8273.2	T-BC/T-GM constant time error classes (A 50 ns, B 20 ns, C 10 ns) and holdover/wander masks
ITU-T G.8272	PRTC-A 100 ns / PRTC-B 40 ns accuracy
ITU-T G.8260	Definitions and measurement methodology for packet-timing metrics — the basis for the instrument floor in §8
ITU-T G.8262 / G.8264 / G.8261	SyncE EEC, ESMC/SSM quality levels, and wander limits (catalogued; not exercisable here — see §10)
O-RAN.WG4.CUS.0	Fronthaul LLS-C1..C4 sync configurations; cluster TAE budget of 130 ns (±65 ns)
O-RAN.WG11 Threat Modeling (ETSI TR 104 106 V3.0.0)	S-plane threat identifiers T-SPLANE-01...04 mapped to scenarios in §9
linuxptp v4.0	Reference implementation; pmc management datasets, PORT_STATS_NP and PORT_SERVICE_STATS_NP counters
TIMESAFE (ACM ToPS 2025, doi:10.1145/3775060)	Real fronthaul hardware captures used to verify profile agreement, and the source of the attack behaviour modelled

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